Notice, as well, a couple of key features of quantum teleportation. Since Photon A's original quantum state was disrupted by my measurement, Photon C in London is now the only one in that original state. There aren't two copies of the original Photon A and so, rather than calling this quantum faxing, it is indeed more accurate to call this quantum teleportation.[4] Furthermore, even though we teleported Photon A from New York to London—even though the photon in London becomes indistinguishable from the original photon we had in New York—we do not learn Photon A's quantum state. The photon in London has exactly the same probability of spinning in one direction or another as Photon A did before my meddling, but we do not know what that probability is. In fact, that's the trick underlying quantum teleportation. The disruption caused by measurement prevents us from determining Photon A's quantum state, but in the approach described, *we don't need to know the photon's quantumstate in order to teleport it.* We need to know only an aspect of its quantum state—what we learn from the joint measurement with Photon B. Quantum entanglement with distant Photon C fills in the rest.

Implementing this strategy for quantum teleportation was no small feat. By the early 1990s, creating an entangled pair of photons was a standard procedure, but carrying out a joint measurement of two photons (the joint measurement on Photons A and B described above, technically called a *Bell-state measurement*) had never been attained. The achievement of both Zeilinger's and De Martini's groups was to invent ingenious experimental techniques for the joint measurement and to realize them in the laboratory.[5] By 1997 they had achieved this goal, becoming the first groups to achieve the teleportation of a single particle.

Realistic Teleportation

Since you and I and a DeLorean and everything else are composed of many particles, the natural next step is to imagine applying quantum teleportation to such large collections of particles, allowing us to "beam" macroscopic objects from one place to another. But the leap from teleporting a single particle to teleporting a macroscopic collection of particles is staggering, and enormously far beyond what researchers can now accomplish and what many leaders in the field imagine achieving